Mahdi Salem,

Senior Data Engineer, Data Architect

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Turning operational complexity and data into insightful, controlled, decision ready intelligence.

Professional Profile

A Microsoft Certified Senior Data Engineer and Data Architect with over 10 years of experience designing data integration, warehousing, and analytics solutions, including eight years supporting enterprise-scale aerospace manufacturing at Bombardier. Holds the **Microsoft Fabric Data Engineering Associate (DP-700)** certification with hands-on experience architecting medallion lakehouse pipelines, Direct Lake semantic models, and end-to-end data platform modernization. Proven track record developing and maintaining enterprise SSIS packages across multi-factory, multi-country operations, integrating operational data from SAP S/4HANA, MES, and legacy databases, and establishing data-quality frameworks that deliver consistent, governed information to analytics consumers. Combines deep technical engineering with business analysis and team leadership to transform complex operational data into trusted, decision-ready assets.

Experienced across legacy on-premises data warehouses — including Oracle 10g GBS and SQL Server — and modern cloud platforms spanning Azure Synapse Analytics, Databricks, and Microsoft Fabric. Led the migration and ingestion of 20+ enterprise data sources into Azure Synapse and Databricks, and subsequently architected the progression of 35+ sources into Microsoft Fabric using Data Pipeline and Dataflow Gen2, delivering curated tables and datamarts through medallion architecture for enterprise analytics.

Core Competencies

Enterprise Data Architecture: Medallion lakehouse architecture, Data Vault 2.0, Kimball dimensional modeling, star schema design, and semantic layer strategy for transactional-to-analytical data progression.

Microsoft Fabric & Cloud Platforms: OneLake, Lakehouse, Warehouse, Data Factory, Dataflow Gen2, Direct Lake, Spark; Azure Synapse Analytics, Databricks (Delta Lake, Unity Catalog), PySpark.

Data Pipeline Engineering: End-to-end ETL/ELT design, ingestion contracts, transformation frameworks, and serving-layer patterns using SSIS, Fabric Data Factory, Databricks, Kafka, and dbt.

Legacy Migration & Modernization: SQL Server, SSIS, SSAS, and SSRS migration to Fabric equivalents; wave planning, parity validation, and legacy decommissioning in enterprise environments.

Data Quality & Governance: KPI-driven data quality frameworks, data reconciliation (SAP/MES/operational), Microsoft Purview catalog design, lineage, RLS/OLS, data masking, and data classification.

SQL & Analytics Engineering: Expert T-SQL and PL/SQL; PySpark transformations; Power BI (Direct Lake, semantic models, DAX), and analytics serving layers.

Technical Leadership: Cross-functional data team delivery, engineering mentorship, architectural decision documentation with trade-offs, stakeholder alignment, and reusable standards development.

Professional Experience

Lead Data Engineer & BI Architect **Aug 2019 – Sep 2026**

Bombardier Aerospace *Mississauga, ON*

- Architected and executed the migration of 20+ enterprise on-prem data sources — including SAP S/4HANA, MES, and legacy production databases - first into Azure Synapse Analytics and then to azure Databricks, establishing medallion-layer transformations with Delta Lake and PySpark for standardized, analytics-ready data delivery.
- Led the data platform progression into Microsoft Fabric, onboarding 15+ data sources via Data Pipeline and

20+ via Dataflow Gen2 into OneLake; designed lakehouse ingestion patterns, curated transformation layers, and Direct Lake–enabled datamarts serving enterprise analytics and reporting.

- Defined and implemented data governance across the platform, including RLS/OLS access policies, data masking, metadata management, and data classification practices to ensure consistent security and compliance across reporting assets both on on-prem and cloud level (purview or Unity Catalouge).
- Designed, developed, and maintained enterprise-grade SSIS packages across a multi-factory, multi-country footprint, improving data-flow accuracy and pipeline efficiency between core operational systems.
- Initiated and implemented a Data Quality KPI framework reconciling SAP, MES, and local production databases, materially improving shop-floor data integrity and establishing shared metric definitions across business units.
- Engineered and deployed over 150 Power BI reports, dashboards, and PowerApps solutions supporting Production Andon teams, Quality Assurance, and PMO governance — establishing standardized semantic models, reusable report templates, and mock-up frameworks under the Bombardier Operational Excellence (BOE) initiative.
- Designed and launched an adaptive OSW (Outstanding Work) Production Tracking web application (.NET / Angular) to structure highly volatile, non-standard assembly tasks, with an integrated attendance management system (Python, Django, SQL).
- Led a team of up to 4 developers and data engineers through detailed planning and role clarity; coached and mentored team members, tailoring guidance and development opportunities to build independent engineering capability.
- Automated cross-departmental operational workflows through scripted tooling and 10+ PowerApps solutions, eliminating manual data entry and accelerating process response times across production and quality teams.

Senior Data and Process Analyst **Jul 2018 – Aug 2019**

Triumph Aerospace Group *Toronto, ON*

- Architected a multi-tiered reporting framework covering daily, weekly, and monthly operational metrics at both macro and aircraft-structural levels, delivering data-driven visibility to line and executive managers.
- Built a Python-based web application for daily attendance management with integrated Crew Load analytics to measure shift efficiency and optimize labor allocation.
- Engineered automated Crew Load Chart generation, cutting processing time from roughly 3 hours to 2 minutes and eliminating manual entry errors.
- Developed real-time alerting mechanisms and dashboards delivering threshold notifications to line and executive managers to prevent project delays.

Process Improvement Lead **Aug 2016 – Jun 2018**

AAA Canada (Aerospace & Transportation) *Toronto, ON*

- Built a Manpower Mapping application distinguishing touch vs. non-touch labor, improving shop-floor visibility and reducing non-value-added idle time.
- Architected a Line Logistic Cage tracking system giving real-time status on job kits and workability post-punch-out, preventing downstream assembly bottlenecks.
- Developed an automated Rapid Non-Conformance (RNC) reporting tool that cut generation time to 3 minutes, accelerating quality-containment response.
- Digitized Daily Management Meeting (DMM) reporting into an automated dashboard, lifting reporting

efficiency by 200% while removing manual data entry.

Lead Business Consultant **Mar 2001 – Jun 2015**

Various Companies (Independent) *Tehran, Iran*

- Directed 30+ consulting engagements in strategic planning, business process re-engineering, TQM, and Six Sigma across multiple industries including pharmaceutical, real-estate, retail, banking, and industrial sectors.
- Delivered market studies, feasibility analyses, and business plans used for investment decisions, financing applications, and new-business launches. Prepared more than 100 proposals and RFQs supporting client performance measurement and organizational improvement.

Technical Expertise

Data Engineering & Architecture	Microsoft Fabric (OneLake, Lakehouse, Warehouse, Data Factory, Dataflow Gen2, Direct Lake), Databricks (Delta Lake, Unity Catalog, PySpark), Azure Synapse Analytics, SSIS, Kafka, dbt, ETL/ELT, Medallion Architecture, Kimball Dimensional Modeling, Data Vault 2.0, SCD Type 1/2/3/6
BI & Analytics	Power BI (Direct Lake, Copilot, Semantic Models, RLS/OLS), SSAS, DAX, Power Query (M), Tableau, SAS, Dataiku, Cognos
Data Governance & DevOps	Microsoft Purview, Data Masking (DDM), Data Classification, Git, GitHub, Azure DevOps Pipelines, CI/CD, Bash, PowerShell
Databases & Infrastructure	SQL Server (T-SQL, SSMS, SSIS, SSAS, SSRS), PostgreSQL, MySQL, Oracle (10g/12c, VPD), Azure Cosmos DB, Docker, Linux/Unix, JSON/REST APIs
Programming & Data Science	Python (PySpark, Pandas, NumPy, SciPy, scikit-learn, Django, Flask), T-SQL, PL/SQL, KQL, R, Jinja2
Application Development	React, Angular, TypeScript, ASP.NET Core, Node.js/Express, HTML5/CSS3, PowerApps, Power Automate

Education & Training

Master of Business Management, Marketing

Allameh Tabatabaie University, 1999 – 2002

Bachelor's Degree, Business Management

Allameh Tabatabaie University, 1996 – 1999

Professional Training

- Business Analytics; Duke University, 2016 – 2017
- CS50's Web Programming with Python and JavaScript; Harvard University, 2020
- Microsoft Power Platform (Power BI, PowerApps, Power Automate); 2020 – 2021

Certifications

Microsoft Certified:

- Data Engineering in Fabric Associate (DP-700)
- Fabric Analytics Engineer Associate (DP-600)
- Power BI Data Analyst Associate (PL-300)
- Power Platform App Maker Associate (PL-100)
- Power Platform Fundamentals (PL-900)
- Azure Data Fundamentals (DP-900)
- Excel 2016 Expert

- Certified ScrumMaster I (CSM)

Six Microsoft badges held in total; some pending renewal.